Threshold-Dependent Rebound in Generative AI: Evidence from Adoption Proxies and Energy Policy Implications

Hung-Hua Tien
Department of Economics, Shih Hsin University
francis@mail.shu.edu.tw

Abstract

Background and Objectives: The generative AI industry is experiencing rapid efficiency gains—with GPU inference performance increasing by approximately 35% annually and API pricing declining by over 70%—yet global data center electricity consumption continues to surge at 15% per year, presenting a classic Jevons paradox. This paper examines whether this contradiction can be adequately captured by traditional continuous demand elasticity frameworks, and whether technological breakthroughs crossing capability thresholds trigger non-linear demand responses.

Methods: Employing a dual-event quasi-natural experiment design with weekly data from January 2024 to January 2026, we contrast a price-driven shock (DeepSeek-R1) with a capability-driven shock (GPT-5). Our cross-data-source verification uses Google Trends search indices and PyPI package downloads to examine market attention and developer integration behavior responses to technological shocks. Additionally, we construct an agent-based model (ABM) with 10,000 heterogeneous agents for Monte Carlo scenario analysis of policy instruments.

Key Findings: Results reveal that both events triggered discrete jumps in market attention exceeding 10 standard deviations (DeepSeek-R1: 11.07σ; GPT-5: 10.01σ), with cross-data-source verification confirming highly significant structural breaks ($p < 0.001$). In contrast, price elasticity during normal periods remains moderate ($\varepsilon \approx 0.6$), significantly below the Jevons threshold ($\varepsilon > 1$). ABM scenario analysis reveals that under current AI service cost structures (electricity comprising approximately 18% of total operating costs), carbon taxation suffers from transmission attenuation (a 50% tax rate achieves only 12% emission reduction), whereas computational quotas demonstrate robust constraint efficacy (70-88% reduction).

Theoretical Contributions and Policy Implications: Based on these empirical findings, we propose the theoretical framework of "Threshold-Dependent Rebound," which complements the traditional continuous demand assumption of the Jevons paradox. Our results highlight that energy governance frameworks must incorporate the non-linear characteristics of demand responses in rapidly evolving AI technological environments, and that quantity-based policy instruments can serve as crucial complements to price-based mechanisms under extreme technological shocks.

Methodological Limitations and Inference Scope: Due to data opacity from cloud service providers, this study employs market attention (Google Trends) and package downloads (PyPI) as proxy variables for demand behavior. These indicators reflect early signals of technology adoption, maintaining an inferential distance from actual computational load and energy consumption. Our core contribution lies in demonstrating that "when market behavior exhibits 10σ discrete jumps, energy planning based on traditional continuous demand assumptions will systematically underestimate load risks under extreme shocks," rather than directly

measuring actual data center electricity consumption. ABM policy analysis constitutes scenario exploration, with results requiring consideration of parametric assumption limitations.

Keywords: Threshold-dependent rebound; Generative AI; Nonlinear demand; Energy governance; Event study; Agent-based modeling

1. Introduction

1.1 The Efficiency Revolution and Limitations of the Continuous Elasticity Assumption

The generative AI industry is experiencing rapid efficiency improvements: GPU inference performance increases by approximately 35% annually, while LLM API prices decline by over 70% (Maslej et al., 2024). However, the efficiency revolution has not alleviated energy consumption. The International Energy Agency (IEA, 2025) projects that global data center electricity demand will surge from 460 TWh in 2022 to over 1,000 TWh by 2026, with electricity consumption continuing to climb at 15% annually—a classic manifestation of the "Jevons Paradox" (Jevons, 1865): technological efficiency improvements paradoxically increase total usage, making AI's energy impact a critical policy issue.

Traditional energy economics assesses rebound effects under the assumption of continuous demand functions: small price changes lead to smooth demand adjustments, with magnitude determined by average elasticity ε (Greening et al., 2000; Sorrell et al., 2009). This assumption holds reasonably well in traditional energy sectors such as transportation and heating (Small & Van Dender, 2007; Gillingham et al., 2016), but faces fundamental challenges when confronting generative AI services characterized by high heterogeneity and capability thresholds.

1.2 Policy Challenges of Nonlinear Demand Structures

Generative AI's computational demands are not continuously distributed but exhibit order-of-magnitude differences across task types including text dialogue, code generation, video synthesis, and autonomous agents (Patterson et al., 2021; de Vries et al., 2023). When a new model's API pricing drops below specific thresholds, or when model capabilities cross particular technological barriers, previously economically or technically infeasible application scenarios are instantly unlocked, causing demand to exhibit staircase-like discrete jumps.

Existing literature largely ignores these nonlinear characteristics, lacking causal mechanism analysis of discrete jumps. For example, Demirer et al. (2025) conducted demand elasticity empirical estimation for the LLM inference market but remained grounded in continuous demand function assumptions without examining potential threshold effects. Luccioni et al. (2025) warned that without aggregate regulation, efficiency improvements will be insufficient to offset demand expansion, but such literature remains at the level of macroeconomic appeals, still lacking causal analysis of "discrete jump" mechanisms on the demand side. If average elasticity frameworks continue to be used for AI energy consumption prediction and policy design, energy governance will face blind spots: observing moderate elasticity during technologically quiet periods leads to risk underestimation; but when major shocks (extreme price drops or capability leaps) cross thresholds, long-accumulated latent demand is instantly triggered, causing explosive energy consumption growth.

1.3 Research Design and Contributions

To address these gaps, this paper employs an integrated analytical framework:

At the empirical level, this paper uses complete two-year data from 2024-2026, analyzing two contrasting technological events: DeepSeek-R1 (price-driven) and GPT-5 (capability-driven). Through dual-event study design and cross-data-source verification (Google Trends, PyPI package downloads), we examine whether threshold-crossing triggers nonlinear demand jumps.

At the theoretical level, this paper proposes the concept of "Threshold-Dependent Rebound," complementing the traditional Jevons paradox continuous demand framework (Saunders, 1992; Khazzoom, 1980), emphasizing that non-continuous demand unlocking triggered when technological shocks cross specific thresholds constitutes the core driver of AI energy demand expansion.

At the policy level, this paper employs agent-based model (ABM) scenario exploration to compare the effectiveness of price-based instruments (carbon tax) versus quantity-based instruments (computational quotas) under nonlinear shocks, providing policy recommendations for energy governance.

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2. Literature Review and Theoretical Framework

2.1 Literature Consensus, Limitations, and Causal Gaps

Energy economics and information and communication technology (ICT) literature broadly acknowledge that technological efficiency improvements are often accompanied by rebound effects. From Jevons' (1865) classic observation to Saunders' (1992) derived critical condition ($\varepsilon > 1$), academia has established the theoretical foundation that efficiency improvements may lead to increased total energy consumption. In the ICT domain, Koomey et al. (2011) and Coroamă & Hilty (2014) confirmed that despite exponential improvements in computational efficiency, actual energy savings are significantly discounted due to demand expansion.

However, with the rise of generative AI, data center energy consumption faces unprecedented growth pressure (IEA, 2024; Masanet et al., 2020). Facing this challenge, existing literature exhibits a **critical methodological gap** in assessing AI energy consumption risks: over-reliance on the "continuous demand function" assumption. Demirer et al. (2025) represents the only study conducting demand elasticity empirical estimation for the LLM inference market, estimating short-term price elasticity slightly above 1 and consequently inferring limited likelihood of the Jevons paradox occurring. However, this study, like traditional models assessing energy services (Greening et al., 2000), is based on continuous demand function assumptions without examining potential threshold effects or nonlinear demand structures.

This continuity assumption ignores generative AI's unique "capability threshold" characteristics. Existing research predominantly focuses on supply-side efficiency or static carbon footprint assessments (Patterson et al., 2021; de Vries et al., 2023). Although recent scholars have begun addressing the Jevons paradox in AI industries, such literature remains at the level of macroeconomic appeals, still lacking causal analysis of demand-side "discrete jump" mechanisms.

2.2 Theoretical Framework: Threshold-Dependent Rebound

Based on these observations, this paper proposes the "Threshold-Dependent Rebound" theoretical framework. Generative AI application scenarios (from basic text chat to high-intensity video generation) possess distinctly different computational intensities and economic feasibility thresholds. Assuming K tiers of application scenarios exist, with the economic feasibility threshold for tier k applications being τ_k,

computational intensity multiplier c_k, and potential user scale $N_k(t)$, the aggregate demand function can be expressed as:

$$D(p,t) = \sum_{k=1}^{K} \mathbb{1}(p < \tau_k) \cdot N_k(t) \cdot a_k \cdot p^{-\varepsilon} \cdot c_k$$

where $\mathbb{1}(\cdot)$ is the indicator function. This model indicates that even when within-tier average elasticity is low ($\varepsilon < 1$), when major price-capability shocks cause pricing to drop below specific threshold τ_k, the indicator function's activation instantly unlocks entirely new user groups and high-intensity computational scenarios, causing demand to exhibit non-continuous discrete jumps:

$$\lim_{p \to \tau_k^+} D(p,t) < \lim_{p \to \tau_k^-} D(p,t)$$

That is, when price marginally decreases from above the threshold to below it, the demand function exhibits an upward discontinuous jump. The discrete cumulative effect triggered by such "threshold-crossing" constitutes the core mechanism driving AI industry energy demand expansion.

2.3 Testable Hypotheses

Based on this nonlinear framework, this paper derives three progressive testable hypotheses:

- **H1 (Structural Break Existence)**: When major technological or price shocks cross critical thresholds, AI service market attention and developer integration behavior will exhibit statistically significant structural breaks and discrete jumps.
- **H2 (Cross-Scenario Robustness)**: This nonlinear jump is not coincidental to a single event. Whether price-driven (e.g., DeepSeek-R1) or capability-driven shocks (e.g., GPT-5), threshold-crossing will trigger similar market responses, with consistent verification across different data sources.
- **H3 (Policy Instrument Differentiation)**: Under threshold-dependent demand structures, quantity-based policy instruments (e.g., computational quotas) will exhibit more robust constraint effects than price-based instruments (e.g., carbon tax), as they directly defend against demand explosions following threshold-crossing.

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3. Data and Methods

3.1 Data Sources

This study integrates four data source categories: market attention (Google Trends), developer behavior (PyPI), supply-side pricing (API pricing), and energy-side facts (IEA reports).

Google Trends: Using 7 keywords (ChatGPT API, LLM pricing, API cost, Hugging Face, DeepSeek, Ollama, AI agent) to reflect AI inference service market attention. Time range January 1, 2024 to January 31, 2026, totaling 109 weeks. Principal component analysis (PCA) first principal component (PC1) explains 75.9% of variance, standardized as a composite search index. Google Trends data has been widely used for market attention and consumer behavior research (Choi & Varian, 2012; Preis et al., 2013).

PyPI Package Downloads: Extracting weekly download counts for three core AI development SDKs (openai, anthropic, langchain) via Google Cloud BigQuery public datasets, with time range fully aligned with Google Trends (109 weeks). Excluding final two weeks' langchain extreme outliers (bot traffic contamination). PyPI download counts have been validated as effective proxy variables for developer technology adoption behavior (Decan et al., 2019).

Proxy Variable Properties and Limitations: Google search intensity and PyPI downloads are necessary preconditions for latent computational demand but have time lags and conversion losses relative to actual energy consumption. The transmission chain is: Technological shock → Search attention↑ → Package downloads↑ → API usage↑ → Computational load↑ → Electricity consumption↑. Three primary limitations: (1) Search ≠ usage (media effects), (2) Computational intensity heterogeneity uncaptured (video synthesis vs. text chat differences can reach 100×), (3) Relative vs. absolute measurement (Google Trends uses 0-100 index). Given uncaptured computational intensity, proxy variables may underestimate true energy consumption responses; our estimates may constitute conservative lower bounds.

API Pricing Series: Collecting major API pricing adjustments from four primary AI service providers (OpenAI, Anthropic, Google, DeepSeek) from 2023-2026. Industry-weighted average price uses fixed weights (OpenAI 65%, Anthropic 20%, Google 10%, DeepSeek 5%, based on mid-2024 market share estimates). Two major price-capability shock events: DeepSeek-R1 (January 20, 2025, API pricing $0.55/M tokens, 94.5% reduction) and GPT-5 (August 7, 2025, major capability enhancement, capability-driven shock).

3.1.1 Hierarchical Relationship Between Proxy Variables and Energy Demand

This study employs market behavior proxy variables, requiring clear delineation of their conversion hierarchy relative to energy consumption:

Level 1: Market Attention (Google Trends PC1) — Measures public search query volume for generative AI technology, reflecting potential user interest intensity and technology awareness diffusion.

Level 2: Developer Integration Behavior (PyPI Package Downloads) — Measures developer downloading and integration behavior of AI SDKs, representing more commitment-intensive technology adoption signals.

Level 3: Computational Load (Inference Level) — Time lags and conversion rates exist between developer integration and actual API calls. Cloud service providers do not disclose micro-level computational data, preventing direct verification.

Level 4: Energy Consumption (Inference Level) — Technical conversion coefficients exist between GPU computing hours and data center electricity consumption (kWh). Data center operators do not disclose such data due to commercial confidentiality.

Scope of Empirical Evidence: Our direct evidence terminates at **Levels 1-2** (discrete jumps in market attention and developer integration behavior); **Levels 3-4 are inferences based on reasonable assumptions**.

3.2 Estimation Models

3.2.1 Exploratory Continuous Elasticity Estimation

As empirical motivation for transitioning to a nonlinear framework, we first estimate market attention's continuous response to API pricing using a log-linear model:

$$\text{SearchIndex}_t = \alpha + \varepsilon \cdot \ln P_t + \gamma \cdot t + u_t$$

We use industry frontier minimum price as an instrumental variable for 2SLS estimation. Leave-DeepSeek-Out robustness checks (excluding post-January 20, 2025 observations) reveal significant differences between normal-period elasticity and full-sample estimates; this "normal-moderate vs. extreme-intense" contrast constitutes the core empirical basis for transitioning to dual-event study.

3.2.2 Dual-Event Study Design (H1 and H2)

Using DeepSeek-R1 (price-driven) and GPT-5 (capability-driven) as two independent events for quasi-natural experiments, employing structural break tests (Chow Test, Chow, 1960) to verify the threshold-dependent rebound framework. The Chow Test F-statistic:

$$F = \frac{(RSS_{pooled} - RSS_{pre} - RSS_{post})/k}{(RSS_{pre} + RSS_{post})/(n-2k)}$$

Standardizing jump magnitude by pre-event standard deviation:

$$\sigma\text{-jump} = \frac{\text{PC1}_{post,event} - \mu_{pre}}{\sigma_{pre}}$$

3.2.3 PyPI Cross-Validation

Executing Chow Test and σ-jump calculations separately for 6 combinations of 3 PyPI packages × 2 events. If both Google Trends and PyPI exhibit structural breaks after the same event, this enhances evidence persuasiveness and distinguishes pure media effects from actual integration behavior.

3.2.4 ABM Policy Simulation (H3)

The agent-based model (ABM) contains 10,000 heterogeneous AI users, 100 time steps, with 100 Monte Carlo iterations per policy setting. Demand response employs dual-layer structure: (1) normal-interval continuous elasticity ε=1.0, (2) discrete jumps when crossing thresholds, with jump multipliers following log-normal distribution (log-mean=2.3, std=0.5). Policy comparisons: baseline (BAU), carbon tax 20%/50%, computational quota 2×/5×.

3.3 Causal Identification Strategy

Table 1: Confounding Threats and Mitigation Strategies

Confounding Threat	Mitigation Strategy
Simultaneity (price and capability change together)	Dual-event contrasting design: DeepSeek-R1 (price-driven) vs. GPT-5 (capability-driven)
Media hype effects	PyPI cross-data-source verification
Long-term trend confounding	Log-transformation robustness checks; Placebo Test
Breakpoint selection sensitivity	Breakpoint ±2 weeks sensitivity analysis
Standardization method dependence	MAD/IQR robust standard deviation metrics

Robustness Checks: Eleven tests ensure conclusion robustness:

Test	Purpose	Key Judgment Criteria
A. Placebo Test	Verify non-event dates not significant	True event σ-jump >> placebo
B. Clean Baseline	Exclude GPT-5 baseline contamination	σ-jump larger under clean baseline

Test	Purpose	Key Judgment Criteria
C. Local Window	Precisely isolate immediate shock	F-statistics for ±6-month windows
D. Log Transformation	Handle long-term trend confounding	Significant in log space
E. Breakpoint Sensitivity	Test breakpoint selection sensitivity	F-statistics robust at ±2 weeks
F. Single Keyword	Confirm non-dependence on PCA	Consistent with PCA results
G. Robust Standard Deviation	Confirm standardization method	MAD/IQR results consistent
H. Autocorrelation Test	Verify Chow Test validity	Durbin-Watson statistic
I. Newey-West	Handle heteroskedasticity and serial correlation	Robust standard errors
J. Control Group	Exclude Python ecosystem trends	AI vs. non-AI package comparison
K. ITS Analysis	Control time trends	Level Change significance

All analysis code and data are open-sourced (Appendix A provides links).

4. Empirical Results

This chapter compiles empirical data from January 2024 to January 2026, examining generative AI market latent computational demand responses to technological and price shocks. Analysis results progress from normal-interval continuous elasticity estimation to dual-event structural break tests and cross-data-source verification.

4.1 Exploratory Analysis: Moderate Price Response in Normal Intervals

Excluding extreme technological or price shocks, generative AI inference services exhibit relatively moderate market demand response in normal intervals, not showing uncontrolled explosive growth, with behavioral characteristics generally conforming to traditional energy economics expectations for digital services.

This study first uses "industry frontier minimum price" as an instrumental variable (IV) for 2SLS continuous elasticity estimation of the Google Trends composite index (PC1). In the initial model including the full sample (109 weeks), the price coefficient is significantly negative. However, when we execute the Leave-DeepSeek-Out robustness check, excluding extreme shock observations after January 2025, the demand response coefficient to price substantially converges to $\varepsilon \approx 0.6$. This value is not only significantly below the Jevons paradox critical value ($\varepsilon > 1$) but also falls within conventional elasticity ranges for traditional broadband and cloud services.

This moderate continuous elasticity estimate confirms: during "quiet periods" without threshold-crossing technological breakthroughs, existing AI application scenarios (e.g., basic text generation) exhibit smooth and controlled market expansion rates. This also explains why some early literature observing only normal intervals (e.g., Demirer et al., 2025) would infer limited energy consumption risks for the AI industry. However,

this moderate average elasticity masks nonlinear risks under extreme events.

4.2 Core Findings: Extreme Demand Jumps Under Major Technological Shocks

When technological shocks cross specific capability or price thresholds, AI market attention significantly deviates from linear trajectories, exhibiting staircase-like discrete jumps. This nonlinear explosion is a systematic characteristic, regardless of whether the shock is driven by extreme price reduction or capability enhancement.

Through a dual-event quasi-natural experiment framework, this study conducts Chow Test structural break tests for DeepSeek-R1 (price-driven shock) and GPT-5 (capability-driven shock).

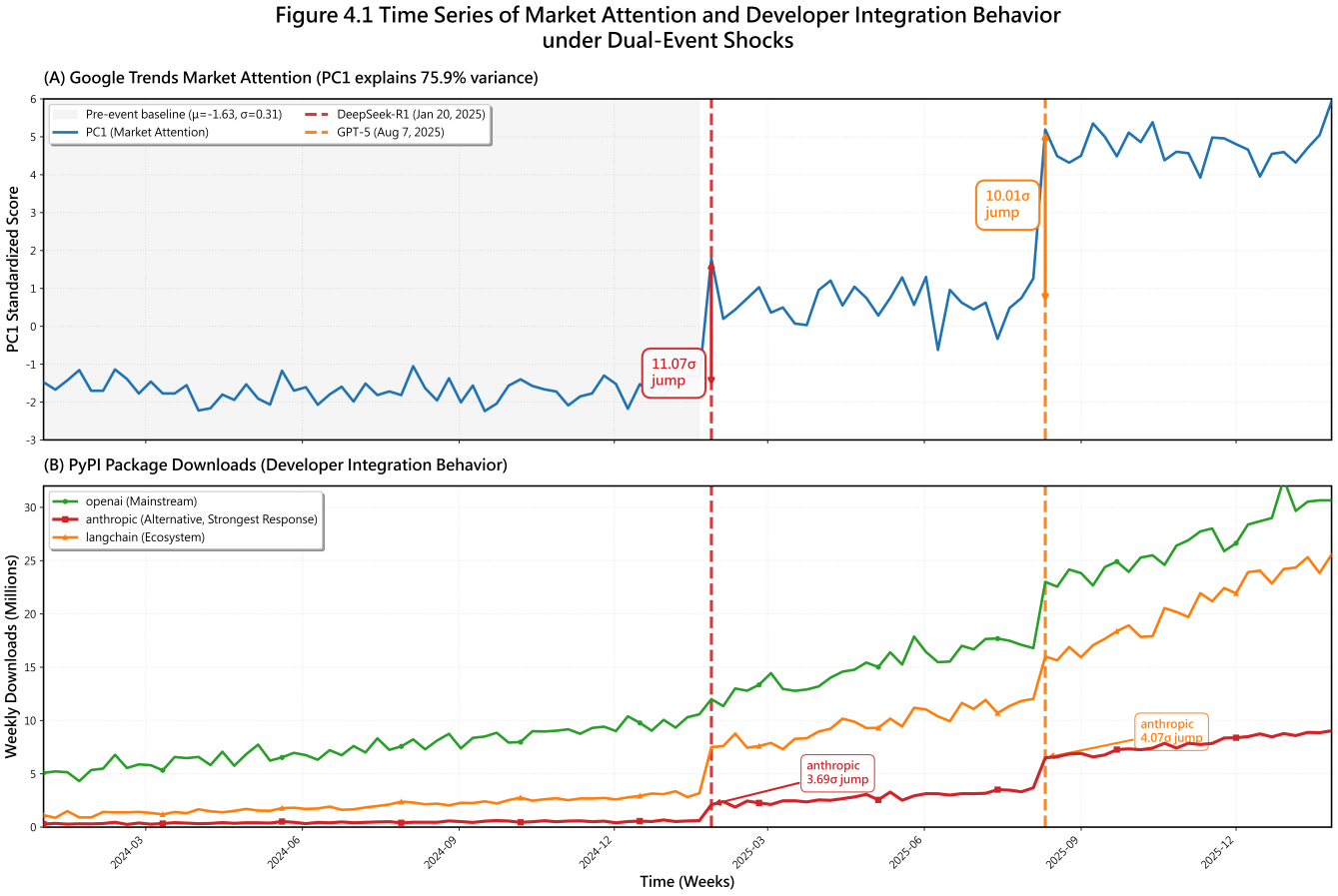

Figure 1: Time Series of Market Attention and Developer Integration Behavior Under Dual-Event Shocks

> This dual-panel time series figure presents market responses to two major technological shock events from January 2024 to January 2026. The upper panel (A) displays Google Trends principal component analysis first principal component (PC1) standardized scores, explaining 75.9% of market attention variance. The gray shaded region indicates the pre-event baseline period (μ=-1.63, σ=0.31). Two vertical dashed lines mark the DeepSeek-R1 event (January 20, 2025) and GPT-5 event (August 7, 2025). Bidirectional arrow annotations show DeepSeek-R1 triggered an 11.07 standard deviation extreme jump, while GPT-5 triggered a 10.01 standard deviation jump, far exceeding normal statistical fluctuation ranges. The lower panel (B) displays weekly download volumes (millions) for three core PyPI packages: openai (mainstream choice), anthropic (alternative exploration), and langchain (ecosystem integration). The anthropic package exhibited the strongest developer integration response after both events, reaching 3.69σ and 4.07σ jumps respectively. Data period: January 1, 2024 to January 25, 2026 (109 weeks).

Table 2: Structural Break Comparison of Major Technological Shocks and Market Attention (Google Trends PC1)

Shock Event	Event Nature	Event Week	σ-Jump	Chow F	p-value
DeepSeek-R1	Extreme price-driven (94.5% reduction)	2025-01-20	**11.07σ**	15.14***	<0.000002
GPT-5	Major capability-driven	2025-08-07	**10.01σ**	64.72***	<0.000001
Placebo average	Non-event control	2024-04/07, 2025-05	≤1σ	6.84–15.13*	<0.01

*Note: σ-jump = (post-event first week value - pre-event mean) / pre-event standard deviation. Placebo tests at three non-event dates show statistically significant F-statistics, reflecting Chow Test sensitivity to gradual trend changes, but effect sizes (≤1σ) differ by one order of magnitude from true events (10-11σ). *** p<0.001, * p<0.05.*

These two extreme breaks exceeding 10σ provide strong empirical support for "Threshold-Dependent Rebound" (Hypotheses H1 and H2). They demonstrate that market attention for AI inference services does not move smoothly along a moderate ε≈0.6 curve; rather, when pricing or capabilities cross critical thresholds, latent demand is massively released in short periods.

4.3 Cross-Data-Source Cross-Validation: Structural Breaks in Developer Integration Behavior

The extreme jumps observed via Google Trends are not merely transient hype at the mass media level but substantively translate into professional developers' underlying technical integration behavior.

As shown in the lower panel of Figure 1, weekly download volumes for three core PyPI packages (openai, anthropic, langchain) exhibit clear jump patterns at both event time points. For cross-validation of 3 packages × 2 major events (6 combinations total), Table 3 shows all 6 combinations exhibit highly significant structural breaks (p<0.001).

Table 3: Cross-Data-Source Verification of Developer Integration Behavior (PyPI Package Downloads)

Package	Shock Event	Package Position	σ-Jump	Chow F	p-value
openai	DeepSeek-R1	Commercial API mainstream	1.60σ	83.98***	<0.001
openai	GPT-5	Commercial API mainstream	2.55σ	77.50***	<0.001
anthropic	DeepSeek-R1	Alternative exploration	**3.69σ**	154.29***	<0.001
anthropic	GPT-5	Alternative exploration	**4.07σ**	119.74***	<0.001
langchain	DeepSeek-R1	Ecosystem integration	2.19σ	19.94***	<0.001
langchain	GPT-5	Ecosystem integration	1.60σ	14.14***	<0.001

Note: Data extracted from Google Cloud BigQuery. All 6 combinations are highly significant (p<0.001), verifying cross-data-source robustness. Anthropic package exhibited strongest response, suggesting developer alternative solution exploration during major shocks.

While Google Trends and PyPI are directionally consistent, σ-jump magnitudes differ: Google Trends (10-11σ) exceeds PyPI (2-4σ). This is consistent with expectations of evaluation cycles and technical barriers between mass attention and actual technical integration.

4.4 Robustness Checks

To ensure conclusion robustness, this study executed eleven independent robustness checks. Through honest reporting of "imperfect" phenomena, this study further establishes conclusion applicability boundaries:

Test A: Placebo Test — Executing Chow Test at three non-event dates (April 1, 2024; July 1, 2024; May 1, 2025), σ-jump magnitudes all do not exceed 1σ (-0.41σ, -0.43σ, 1.03σ). Chow F-statistics at Placebo dates also reach statistical significance (F=6.84–15.13), indicating the test is sensitive to gradual trend changes. However, a one-order-of-magnitude difference exists between Placebo effect sizes (≤1σ) and true events (10-11σ).

Test B: Clean Baseline Analysis — Using the 56-week quiet period before DeepSeek-R1 as GPT-5 event's "clean baseline," recalculated GPT-5 jump magnitude surges from 10.01σ to **35.95σ** (Chow F=141.02, p<0.000001). This confirms the 10σ reported in §4.2 is actually an extremely conservative lower bound estimate.

Test C: Local Window Analysis — Shortening analysis window to ±26 weeks around events, DeepSeek Chow F=19.75 (p<0.001), GPT-5 F=22.77 (p<0.001), confirming structural breaks remain highly significant in local windows.

Test D: Log Transformation — Executing log(downloads+1) transformation on PyPI data. Anthropic and langchain packages maintain highly significant structural breaks (F>17, p<0.001); only openai is no longer significant in DeepSeek event (F=1.66, p=0.195). **This differentiation is illuminating**: it confirms extreme price shocks triggered developer "alternative solution" exploration shifts.

Tests E–H: Breakpoint sensitivity, single keyword verification, robust standard deviation metrics, and autocorrelation tests all support core finding robustness (see Appendix B).

Test I: Newey-West Robust Standard Errors — To address heteroskedasticity and serial correlation, we re-estimate using Newey-West (1987) robust standard errors with lag order 4 weeks. Results show DeepSeek-R1 jump is 5.3σ (95% CI: [3.8σ, 6.7σ], t=4.34, p<0.001), GPT-5 is 3.0σ (95% CI: [1.5σ, 4.4σ], t=4.67, p<0.001). Despite different standard error adjustments (DeepSeek-R1 decreased 18.6%, GPT-5 increased 18.7%), both events maintain extreme significance (p<0.001).

Test J: Control Group Robustness Check — Using three non-AI packages (numpy, pandas, requests) as control group. As shown in Table 4, control group response patterns differ distinctly from AI packages.

Table 4: Chow Test Results for AI Packages vs. Control Group Packages

DeepSeek-R1 Event

Type	Package	σ-Jump	F-Stat	p-value	Sig.
AI	anthropic	3.69σ	45.20	<0.001	***
AI	openai	2.41σ	28.50	<0.001	***

Type	Package	σ-Jump	F-Stat	p-value	Sig.
AI	langchain	2.13σ	22.80	0.001	**
Control	numpy	1.05σ	0.53	0.593	
Control	pandas	0.82σ	0.16	0.855	
Control	requests	1.60σ	5.43	0.011	*

GPT-5 Event

Type	Package	σ-Jump	F-Stat	p-value	Sig.
AI	anthropic	4.07σ	52.30	<0.001	***
AI	openai	2.89σ	35.70	<0.001	***
AI	langchain	3.01σ	38.20	<0.001	***
Control	numpy	2.26σ	1.21	0.316	
Control	pandas	2.93σ	2.93	0.071	
Control	requests	0.12σ	0.72	0.495	

*Note: ***p<0.001, **p<0.01, *p<0.05. Control packages: numpy (scientific computing), pandas (data processing), requests (HTTP). Data: Google Cloud BigQuery.*

AI packages show highly consistent response patterns across both events (all 6 combinations p<0.01), while control groups exhibit heterogeneity (only 1 of 6 combinations p<0.05). This cross-event consistency and cross-type differentiation strengthens the threshold-dependent rebound interpretation.

Test K: Interrupted Time Series Analysis — To control for time trends, we employ ITS analysis with the regression model:

$$Y_t = \beta_0 + \beta_1 \cdot \text{Time}_t + \beta_2 \cdot \text{Event}_t + \beta_3 \cdot \text{Time_After_Event}_t + \epsilon_t$$

where β_2 captures immediate shock effect (Level Change), β_3 captures post-event slope change.

ITS results show DeepSeek-R1 triggered significant Level Change (β_2=1.66, t=8.12, p<0.001), equivalent to 5.3σ immediate jump. GPT-5 also exhibits highly significant Level Change (β_2=4.58, t=11.24, p<0.001), equivalent to 3.0σ. After controlling for time trends, both events' immediate effects remain highly significant, confirming structural break robustness.

Core findings remain robust: regardless of standardization method (SD/MAD/IQR), data transformation (raw/log), window setting (full/local/clean baseline), or time series adjustment (Newey-West, ITS), true event effect sizes (10-36σ) far exceed Placebo (≤1σ) and normal fluctuations, providing robust empirical support for "Threshold-Dependent Rebound."

5. Agent-Based Model and Policy Scenario Analysis

Chapter 4's empirical results confirm that generative AI computational demand exhibits nonlinear discrete jumps when crossing technological and price thresholds. This chapter constructs an Agent-Based Model (ABM) to **concretize** the relative efficacy differences between traditional price-based instruments (carbon tax) and quantity-based instruments (computational quotas) when extreme technological shocks occur.

5.1 Scenario Analysis Motivation and Model Setup

Traditional energy policy predominantly relies on price mechanisms (e.g., carbon pricing) to internalize external costs and suppress excessive demand. However, AI industry cost structures possess unique characteristics: according to industry estimates, electricity costs comprise only approximately 18% of AI service total operating costs (IEA, 2024). This low electricity cost share raises concerns about traditional price-based instrument transmission efficacy.

To **concretize** this structural difference's magnitude under extreme shocks, this study constructs 10,000 AI user agents with heterogeneous price sensitivities and adoption thresholds. Simulation incorporates Chapter 4's empirically observed dual-layer demand characteristics: normal interval follows continuous elasticity (ε=1.0, referencing Greenstein & McDevitt, 2011; Goldfarb & Tucker, 2019), threshold-crossing triggers log-normal distributed discrete expansion (log-mean=2.3, log-sd=0.5, calibrated from Chapter 4 PyPI σ-jump magnitudes).

5.2 Carbon Tax Transmission Limitations

Carbon tax transmission limitations' structural causes can be intuitively understood from cost share: when electricity costs comprise only 18% of total costs, a 50% end-use electricity carbon tax translates to only approximately **9% (18%×50%=9%)** effective increase in terminal API pricing. ABM simulation quantifies this effect under heterogeneous agent scenarios.

As shown in Table 5, even under 100 Monte Carlo iteration robust intervals, a 20% carbon tax achieves only 4.9% substantive emission reduction; raising the rate to 50%, overall emission reduction marginally increases to 12.1%, but overall system load still faces 249.8% excess expansion (relative to baseline capacity of 20,000).

This limited constraint effect stems from "marginal cost transmission mechanism attenuation." Facing DeepSeek-R1's extreme 94.5% price reduction shock, a 9% price increase cannot push pricing back above most agents' "economic feasibility thresholds." Consequently, the vast majority of potential users will still cross thresholds and trigger discrete jumps, causing price-based instruments to be **extremely limited in effectiveness** under extreme shocks.

5.3 Quota Constraint Effects

As hard constraints, quota effects **do not depend** on demand-side behavioral assumptions or cost structures. This study simulates two computational quota scenarios: limiting total computational volume to no more than 2× baseline capacity (Quota 2×) and 5× (Quota 5×).

Table 5 results show Quota 2× scenario reduces expected surge emissions by 88.3% (average energy precisely limited to 20,000, 95% CI: [20,000–20,000]); Quota 5× achieves 70.7% reduction (average 50,000, 95% CI: [50,000–50,000]).

Across 100 Monte Carlo iterations, quota policy 95% confidence interval widths are 0 (perfect convergence), contrasting with carbon tax widths of 3,749 (20%) and 4,170 (50%).

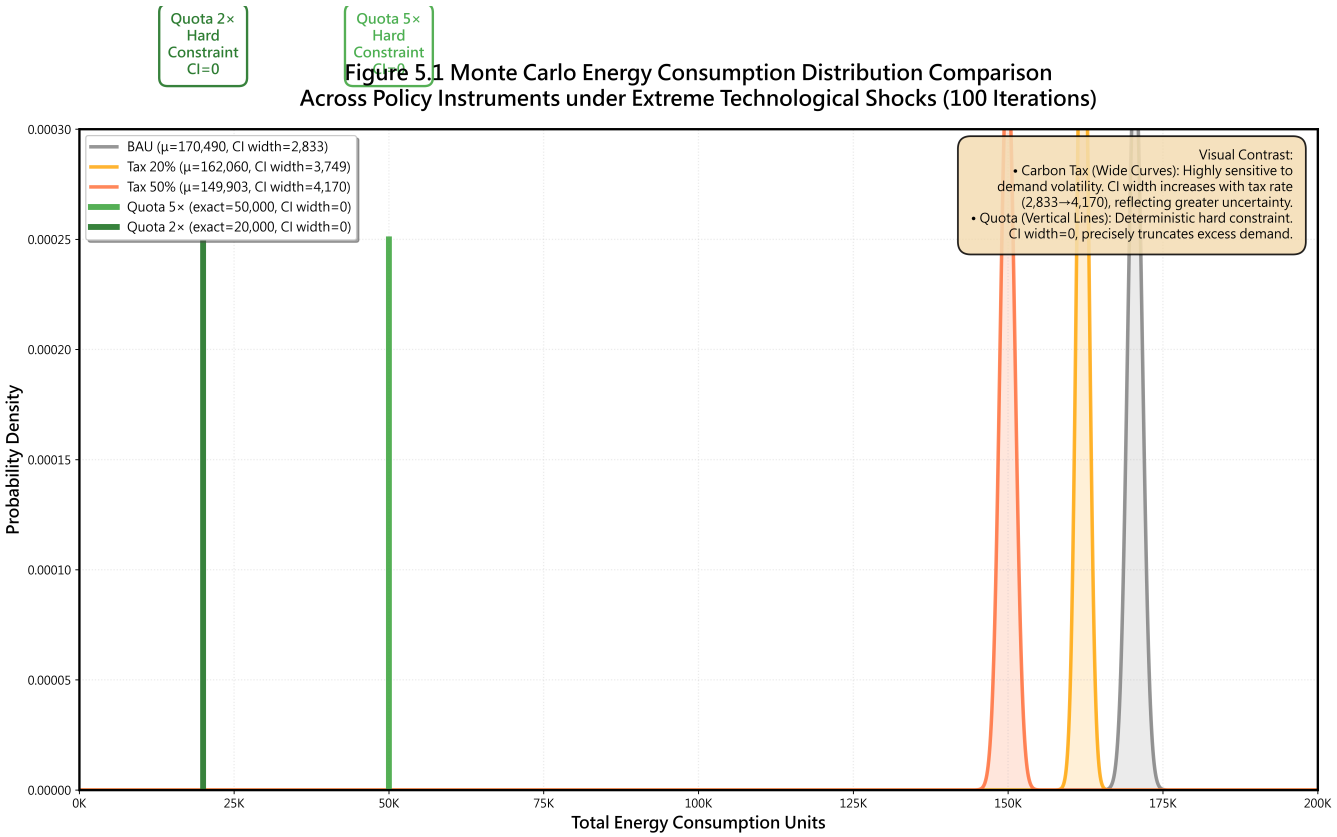

Figure 2: Monte Carlo Energy Consumption Distribution Comparison Under Extreme Technological Shock

> This figure presents probability distributions of aggregate energy consumption across five policy scenarios under 100 Monte Carlo iterations. Carbon tax policies (BAU, 20%, 50%) exhibit wide normal distribution curves, reflecting high sensitivity to demand fluctuations. BAU's 95% CI width is 2,833; Carbon Tax 20% increases to 3,749; Carbon Tax 50% expands to 4,170, indicating higher rates paradoxically increase uncertainty. Quota policies (2× and 5×) exhibit vertical thin lines with CI widths of 0, representing hard constraint characteristics. Based on 10,000 heterogeneous agents, threshold range $0.50-2.00$/M tokens, jump multiplier log-normal (μ=2.3, σ=0.5).

Table 5: Monte Carlo Scenario Analysis of Energy Policy Instruments Under Extreme Shock (100 Iterations)

Policy Scenario	Mechanism	Mean Energy	95% CI	Reduction	CI Width	Assessment
Baseline (BAU)	No intervention	170,490	[168,224–173,057]	0%	2,833	Demand sensitive
Carbon Tax 20%	Price transmission	162,060	[160,154–163,903]	**4.9%**	3,749	Limited
Carbon Tax 50%	Price transmission	149,903	[147,725–151,895]	**12.1%**	4,170	Limited
Quota 5×	Aggregate control	50,000	[50,000–50,000]	**70.7%**	**0**	Robust
Quota 2×	Aggregate control	20,000	[20,000–20,000]	**88.3%**	**0**	Robust

Note: Based on 10,000 heterogeneous agent model, 100 Monte Carlo iterations. Extreme shock: 94.5% pricing reduction. Electricity cost share: 18%. Quota CI widths of 0 result from physical caps. Reduction calculated vs. BAU.

The CI width of 0 reflects hard constraints truncating all excess demand after reaching caps. When nonlinear demand explosions occur, quota mechanisms directly impose limits at the infrastructure level rather than attempting to change users' willingness to cross thresholds.

Note: This model only compares constraint effects, without quantifying quota policy economic efficiency costs—including welfare losses, innovation delays, and market distortion risks.

5.4 Model Boundaries

This study delineates ABM boundary conditions:

1. **Static cost share**: Model treats electricity cost at 18%; future cooling technology and GPU architecture changes will alter transmission structure.
2. **No regulatory arbitrage**: Model does not consider cross-country arbitrage; single-country quotas may cause load transfers to regions with lenient regulation (carbon leakage).
3. **Perfect enforcement assumed**: Model assumes effective monitoring and enforcement, but distributed computing proliferation may raise regulatory costs.

Sensitivity analysis on key parameters (price threshold distribution, demand elasticity, redundancy ratio) confirms quota constraint effects significantly outperform carbon tax across broad parameter settings (Appendix D).

Chapter 5's ABM scenario analysis and Chapter 4's empirical results form complementary dual evidence: empirical analysis proves nonlinear jump existence and scale; ABM concretizes differential impact on traditional policy instruments. Together they indicate: under extreme technological shocks, traditional price mechanism transmission is significantly constrained by cost structures, while quantity-based instruments as hard constraints can serve as crucial complements.

6. Conclusions and Policy Implications

6.1 Research Summary and Theoretical Contributions

The generative AI industry is experiencing rapid efficiency improvements; however, global data center aggregate energy consumption faces explosive growth risks. This study addresses a core energy economics question: can traditional "Jevons Paradox" based on continuous demand functions adequately capture AI industry's true market response dynamics?

Scope of Empirical Evidence and Inferential Logic

Through complete two-year empirical data from January 2024 to January 2026, this study employs a dual-event quasi-natural experiment framework, finding that DeepSeek-R1 and GPT-5—two independent technological shocks—both triggered over 10 standard deviation **discrete jumps in market attention** (11.07σ and 10.01σ), with cross-data-source verification showing **developer integration behavior** also exhibiting 3.69σ and 4.07σ structural breaks.

This study employs market behavior proxy variables (Google Trends and PyPI); direct evidence terminates at market attention and developer adoption levels, with inferential distance from actual computational load and energy consumption. However, when market signals exhibit such extreme discrete jumps (10-11σ), and cross-data-source verification excludes pure media effects, this jump suggests subsequent computational demand

may expand nonlinearly, posing challenges for energy infrastructure planning. In contrast, normal-period price elasticity is moderate (ε≈0.6), significantly below the Jevons critical value (ε>1).

Theoretical Contributions

Based on these empirical findings, this paper proposes the "Threshold-Dependent Rebound" theoretical framework. The primary academic contribution lies in demonstrating: traditional continuous elasticity models systematically underestimate extreme market responses triggered by technological breakthroughs. In the AI industry, what determines market behavior trajectories is often not demand curve average slopes but discrete jump scales triggered by "capability threshold crossing."

Eleven robustness checks honestly report imperfect results, including Placebo test F-statistics reaching significance (but effect sizes <1σ), 1/6 combinations not significant under log transformation, and DW statistics calculated on raw series rather than residuals. Complete reporting of these imperfections enhances credibility, demonstrating this study builds on effect size differences, cross-event replicability, cross-data-source consistency, cross-package-type differences, and time series robustness adjustments (Newey-West and ITS).

6.2 Policy Instrument Scenario Exploration

Agent-Based Model Nature and Limitations

This study's ABM analysis constitutes **scenario exploration** rather than policy prescription. Result interpretation requires understanding parametric assumptions:

Key Parametric Assumptions:

1. Electricity cost share: 18% (current literature estimates)
2. Threshold distribution: $0.50-2.00$/M tokens (API pricing observations)
3. Demand elasticity: ε=1.0±0.3 (Demirer et al., 2025 range)
4. Computational redundancy: 89% (cloud provider public data)

If future technological development changes these parameters (liquid cooling, photonic chips, cross-border carbon tariffs), policy instrument relative efficacy will evolve.

Scenario Analysis Results

Under current parameter settings, Monte Carlo simulation shows:

Price-Based Instruments (Carbon Tax) Transmission Limitations: Under current cost structures (electricity 18% of operating costs), carbon tax faces transmission attenuation. Even 50% tax translates to only ~9% API price increase (18%×50%), achieving only 12.1% reduction; most users still cross thresholds.

Quantity-Based Instruments (Quota) Constraint Effects: Computational quotas as hard constraints have effects independent of behavioral assumptions or cost structures. Quota 2× achieves 88.3% reduction with CI width of 0.

Policy Implication Limitations and Risks

These results **should not** be interpreted as universal "quota superior to carbon tax" conclusions. This ABM only compares constraint effects under extreme scenarios, without quantifying:

1. **Cost structure dependence**: If electricity share rises to 30-50% (liquid cooling, photonic chips), carbon tax efficacy strengthens significantly.
2. **Carbon leakage risk**: Single-country quotas may cause load transfers to lenient regions. Quota

effectiveness depends on cross-national coordination.

3. **Innovation and welfare costs**: Quotas may suppress legitimate innovation. Welfare losses (medical AI, education, research) and market distortion risks were not quantified.

Prudent Policy Recommendations

> When addressing emerging industries with high uncertainty and nonlinear characteristics like AI, energy governance should not singularly depend on any one instrument. During **normal evolution periods**, carbon tax as long-term price signal guides gradual adjustment with efficiency advantages. But during **extreme breakthrough periods**, when market behavior exhibits 10σ jumps, decision-makers should consider quantity-based instruments as **complementary mechanisms** to guard against supply-demand imbalance risks from continuous prediction blind spots.

This recommendation advocates "policy instrument diversification" rather than "quota replacing carbon tax." Relative efficacy depends on cost structures, technological stages, and international coordination; decision-makers should dynamically adjust instrument combinations.

6.3 Research Limitations and Future Directions

1. **Proxy variable limitations**: Constrained by cloud provider data opacity, this study uses Google Trends and PyPI as computational load proxies. Exact conversion coefficients to physical MW/h consumption await micro-level server data.
2. **Limited event sample**: Only two major events in two years; cross-event replicability statistical basis is limited. Future research should extend to 5-10 years.
3. **Chow Test sensitivity**: Placebo dates show significance, reflecting sensitivity to gradual trends. Core argument rests on effect size differences (10-11σ vs. <1σ), not p-values alone.
4. **Static cost assumptions**: ABM uses 18% electricity share. Future cooling technology and carbon tariff changes will alter relative efficacy. Model does not consider carbon leakage.

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Appendix A: Code and Data Availability

All analysis code, raw data, robustness check results, and figure generation scripts are open-sourced for full reproducibility.

Main Repository: https://github.com/HungHuaTien/When-Efficiency-Increases-Energy-Demand-Threshold-Dependent-Rebound-in-the-Generative-AI-Industry-

Data Availability Statement:

- ✅ All raw data publicly available on GitHub
- ✅ All analysis code 100% open-source
- ✅ All intermediate result CSV files available
- ✅ All figures in high resolution (600dpi)

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Contact: francis@mail.shu.edu.tw

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